R Code

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# PRACTICAL 2: VISUALIZATION AND IDENTIFICATION OF TIME SERIES COMPONENTS USING R


# Objective:
# To understand the structure of time series data and to visualize
# real time series data using R. The practical also aims to identify
# trend and seasonal variation and to obtain the trend, seasonal,
# and irregular components using time series decomposition.
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# 1. INTRODUCTION TO THE AirPassengers DATASET
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# AirPassengers is an inbuilt R dataset.
# It contains the monthly totals of international airline passengers
# from January 1949 to December 1960.
# The data are recorded in thousands of passengers.

data(AirPassengers)

# Display the complete time series
AirPassengers

# Display basic information about the dataset
class(AirPassengers)
length(AirPassengers)

# Find the starting and ending time of the series
start(AirPassengers)
end(AirPassengers)

# Find the number of observations in one seasonal cycle
frequency(AirPassengers)

# Explanation:
# frequency(AirPassengers) = 12 means that the observations are monthly.
# Therefore, one complete seasonal cycle contains 12 observations.


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# 2. VISUALIZATION OF THE ORIGINAL TIME SERIES
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# Plot the complete time series
plot(AirPassengers,
     main = "Monthly International Airline Passengers",
     xlab = "Year",
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     ylab = "Number of Passengers (in thousands)")

# From the plot, observe:
# 1. The long-term movement of the series (trend).
# 2. Whether a pattern is repeated at regular intervals (seasonality).
# 3. Whether the magnitude of variation changes over time.


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# 3. EXAMINING THE TREND
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# A trend represents the long-term movement in a time series.
# A moving average can be used to smooth short-term fluctuations.

# Compute a 12-month moving average
trend_ma <- filter(AirPassengers,
                   filter = rep(1/12, 12),
                   sides = 2)

# Plot the original series
plot(AirPassengers,
     main = "Air Passengers with Moving Average",
     xlab = "Year",
     ylab = "Number of Passengers (in thousands)")

# Add the moving average to the plot
lines(trend_ma, lwd = 2)

# The smooth line represents the general long-term movement
# of the time series.


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# 4. EXAMINING SEASONAL VARIATION
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# cycle() identifies the position of each observation within
# one seasonal cycle.

months <- cycle(AirPassengers)

# Display the first 12 cycle values
months[1:12]

# Seasonal plot
monthplot(AirPassengers,
          main = "Seasonal Variation in Air Passengers",
          xlab = "Month",
          ylab = "Number of Passengers (in thousands)")

# Boxplot of monthly values
boxplot(AirPassengers ~ cycle(AirPassengers),
        main = "Monthly Variation in Air Passengers",
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        xlab = "Month",
        ylab = "Number of Passengers (in thousands)",
        names = month.abb)

# The boxplot helps compare the distribution of passenger
# numbers across different months.


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# 5. TIME SERIES DECOMPOSITION
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# A time series can be expressed in terms of:
#
# 1. Trend component
# 2. Seasonal component
# 3. Irregular (random) component
#
# Since the seasonal variation in AirPassengers increases as the
# level of the series increases, a multiplicative decomposition
# is considered appropriate.

decomp_air <- decompose(AirPassengers, type = "multiplicative")

# Display the decomposition object
decomp_air

# Plot all components together
plot(decomp_air)

# The four plots represent:
# 1. Observed series
# 2. Trend
# 3. Seasonal component
# 4. Random/irregular component


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# 6. EXTRACTING INDIVIDUAL COMPONENTS
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# Extract the trend component
trend_component <- decomp_air$trend

# Extract the seasonal component
seasonal_component <- decomp_air$seasonal

# Extract the random/irregular component
random_component <- decomp_air$random

# Display the first few values of each component
head(trend_component)
head(seasonal_component)
head(random_component)
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# 7. PLOTTING THE COMPONENTS SEPARATELY
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# Trend component
plot(trend_component,
     main = "Trend Component",
     xlab = "Year",
     ylab = "Trend")

# Seasonal component
plot(seasonal_component,
     main = "Seasonal Component",
     xlab = "Year",
     ylab = "Seasonal Index")

# Random/irregular component
plot(random_component,
     main = "Irregular Component",
     xlab = "Year",
     ylab = "Random Component")


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# 8. ADDITIVE AND MULTIPLICATIVE DECOMPOSITION
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# Additive model:
# Y(t) = T(t) + S(t) + I(t)

decomp_add <- decompose(AirPassengers, type = "additive")

# Plot additive decomposition
plot(decomp_add)

# Multiplicative model:
# Y(t) = T(t) x S(t) x I(t)

decomp_mul <- decompose(AirPassengers, type = "multiplicative")

# Plot multiplicative decomposition
plot(decomp_mul)

# Compare the two decompositions.
#
# In an additive model, the seasonal variation remains approximately
# constant in magnitude.
#
# In a multiplicative model, the magnitude of seasonal variation
# changes with the level of the series.


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# PRACTICE QUESTION
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# Question:
#
# The inbuilt R dataset "co2" contains monthly atmospheric CO2 concentrations.
# Perform the same analysis on the co2 dataset as demonstrated above for the
AirPassengers dataset.

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